STACKS, COHOMOLOGY OF SCHEMES (COHERENT.TEX): LEMMA-TORSION-HOM-EXT

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 6765-6784 of sources/stacks-coherent/pinned-original-source.tex; proof span is lines 6786-6826, the complete author proof (39 body lines).

Editorial note. Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 47 of queue_coherent.json (pb=1715, stmt 6765-6784, proof 6786-6826).

Editorial note. co02 wave: conflict-free remaining accept (old-session co01-co08 build did not cover this label; fresh adjudication 2026-10-14 accepted).

Editorial note. gate_ok: precard_gate RESULT=OK (tag-based, 2026-10-14); builder raw-interval guard false-positives on problem 376 whose public-items coords are in the upstream frame (6811-6867 vs local 7028-7084).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0880: sources/stacks-coherent/pinned-original-source.tex, source lines 6631–6634. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 010I: sources/stacks-coherent/pinned-original-source.tex, source lines 6826–6826. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0882: sources/stacks-coherent/pinned-original-source.tex, source lines 6655–6666. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0889: sources/stacks-coherent/pinned-original-source.tex, source lines 6704–6720. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:coherent.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let X be a Noetherian scheme and let $\mathcal{I} \subset \mathcal{O}_X$ be a quasi-coherent sheaf of ideals. Any object of $Coh(X, \mathcal{I})$ which is annihilated by a power of $\mathcal{I}$ is in the essential image of (0880). Moreover, if $\mathcal{F}$, $\mathcal{G}$ are in $Coh(\mathcal{O}_X)$ and either $\mathcal{F}$ or $\mathcal{G}$ is annihilated by a power of $\mathcal{I}$, then the maps*

$$\begin{array}{ccc} \mathrm{Hom}_X(\mathcal{F}, \mathcal{G}) & & \mathrm{Ext}_X(\mathcal{F}, \mathcal{G}) \\ \downarrow & & \downarrow \\ \mathrm{Hom}_{Coh(X,\mathcal{I})}(\mathcal{F}^\wedge, \mathcal{G}^\wedge) & & \mathrm{Ext}_{Coh(X,\mathcal{I})}(\mathcal{F}^\wedge, \mathcal{G}^\wedge) \end{array}$$

are isomorphisms.

Proof. Suppose $(\mathcal{F}_n)$ is an object of $Coh(X, \mathcal{I})$ which is annihilated by $\mathcal{I}^c$ for some $c \geq 1$. Then $\mathcal{F}_n \to \mathcal{F}_c$ is an isomorphism for $n \geq c$. Hence if we set $\mathcal{F} = \mathcal{F}_c$, then we see that $\mathcal{F}^\wedge \cong (\mathcal{F}_n)$. This proves the first assertion.

Let $\mathcal{F}$, $\mathcal{G}$ be objects of $Coh(\mathcal{O}_X)$ such that either $\mathcal{F}$ or $\mathcal{G}$ is annihilated by $\mathcal{I}^c$ for some $c \geq 1$. Then $\mathcal{H} = \mathcal{H}om_{\mathcal{O}_X}(\mathcal{G}, \mathcal{F})$ is a coherent $\mathcal{O}_X$-module annihilated by $\mathcal{I}^c$. Hence we see that

$$\mathrm{Hom}_X(\mathcal{G}, \mathcal{F}) = H^0(X, \mathcal{H}) = \lim H^0(X, \mathcal{H}/\mathcal{I}^n\mathcal{H}) = \mathrm{Mor}_{Coh(X,\mathcal{I})}(\mathcal{G}^\wedge, \mathcal{F}^\wedge).$$

see Lemma 0882. This proves the statement on homomorphisms.

The notation Ext refers to extensions as defined in Homology, Section 010I. The injectivity of the map on Ext's follows immediately from the bijectivity of the map on Hom's. For surjectivity, assume $\mathcal{F}$ is annihilated by a power of I. Then part (1) of Lemma 0889 shows that given an extension

$$0 \to \mathcal{G}^\wedge \to (\mathcal{E}_n) \to \mathcal{F}^\wedge \to 0$$

in $Coh(U, I\mathcal{O}_U)$ the morphism $\mathcal{G}^\wedge \to (\mathcal{E}_n)$ is isomorphic to $\mathcal{G} \to \mathcal{E}^\wedge$ for some $\mathcal{G} \to \mathcal{E}$ in $Coh(\mathcal{O}_U)$. Similarly in the other case. □